

WEEKLY PROGRESS JOURNAL

Repair Flow – Right-to-Repair Assistant & Reparability Portal

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Course: UCS503P (Software Engineering)

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Week 1 (Aug 3 - Aug 9): Project Ideation & Team Meetings

- Formed the project team for the UCS503P Software Engineering course.
- Held team meetings and discussions to brainstorm project ideas around the right-to-repair movement and the growing e-waste problem.
- Prepared the initial pitch presentation introducing the project idea and problem statement.

Week 2 (Aug 10 - Aug 16): Project Proposal Drafting

- Held further discussions to refine the problem statement and the proposed solution.
- Contributed to drafting the full project proposal, with a focus on scoping the RAG (Retrieval-Augmented Generation) module for laptop repair guidance.

Week 3 (Aug 17 - Aug 23): RAG Module Setup & Design

- Researched RAG pipeline approaches suited to retrieving laptop repair guides and documentation.
- Defined the ingestion and chunking strategy to be used for repair guide documents.

Week 4 (Aug 24 - Aug 30): Ingestion, Chunking & Embedding Pipeline

- Implemented the document ingestion pipeline, including chunking and embedding generation.
- Created two placeholder repair guide documents (hinge repair, battery replacement) to test the pipeline.

Week 5 & Week 6 (Aug 31 - Sep 12): Retrieval, Evaluation Metrics & CI/CD

- Implemented the retrieval module on top of ChromaDB, returning relevant chunks for a given repair query.
- Set up a GitHub Actions CI workflow running ruff linting and pytest to satisfy the proposal's CI/CD and testing commitments.